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PAPER_1100: SCm Phonon-Modulated LQG Area Operator Derivation

Abstract

We derive the Star Cradle Mechanics (SCm) extension of the loop quantum gravity (LQG) area operator, coupling the Barbero-Immirzi parameter γ , the Planck area ℓ_P^2 , the 26-dimensional buoyancy prefactor $S_{26}^{(3)}$, and the 1.25 THz phonon spectral function $\Phi(\omega, \Gamma)$. The resulting operator $\hat{A}_{\text{SCm}}$ extends the standard LQG area spectrum to include phonon-mediated quantum geometry corrections arising from the UQFF buoyancy sector.

1. Introduction

In canonical loop quantum gravity, the area operator on a spin-network state yields the discrete spectrum:

$$A_{\text{LQG}} = 8\pi\gamma\ell_P^2 \sum_i \sqrt{j_i(j_i + 1)}$$

where $\gamma \approx 0.2375$ is the Barbero-Immirzi parameter and j_i are $\text{SU}(2)$ spin labels at graph punctures through the surface.

The UQFF framework introduces 26-dimensional buoyancy layering through $S_{26}^{(3)}([\text{SSq}]) = (1 - [\text{SSq}])^3$ and a phonon spectral coupling at the characteristic 1.25 THz mode.

2. The SCm Area Operator

For a single puncture with spin j :

$$\hat{A}_{\text{SCm}} = 8\pi\gamma\ell_P^2 \sqrt{j(j + 1)} \cdot S_{26}^{(3)}([\text{SSq}]) \cdot \Phi_{1.25 \text{ THz}}(\omega, \Gamma)$$

where the Lorentzian phonon spectral function is:

$$\Phi(\omega, \Gamma) = \frac{1}{\pi} \frac{\Gamma}{(\omega - \omega_0)^2 + \Gamma^2}$$

with $\omega_0 = 1.25 \times 10^{12}$ Hz and linewidth Γ .

3. Area Gap Modification

The minimum area eigenvalue (area gap) at $j = \frac{1}{2}$:

$$A_{\text{gap}}^{\text{LQG}} = 8\pi\gamma\ell_P^2 \sqrt{\frac{3}{4}} \approx 4\pi\sqrt{3}\gamma\ell_P^2$$

The SCm-modified gap:

$$A_{\text{gap}}^{\text{SCm}} = A_{\text{gap}}^{\text{LQG}} \cdot S_{26}^{(3)} \cdot \Phi(\omega, \Gamma)$$

For $[\text{SSq}] = 0.57$:

$$S_{26}^{(3)} = (1 - 0.57)^3 = 0.43^3 \approx 0.0795$$

4. Physical Implications

The phonon modulation introduces: 1. **Linewidth dependence:** The area gap depends on the decoherence width Γ of the 1.25 THz mode 2. **Buoyancy suppression:** The $S_{26}^{(3)} \ll 1$ factor significantly reduces the effective area gap, potentially observable in black hole entropy corrections 3. **Frequency selectivity:** Only geometries resonant near ω_0 receive maximal area eigenvalues

5. Implementation

Calculator: `SCmLQGAreaOperatorDerivationCalculator` in `CondensedPhysics.py`

- Accepts spin j , frequency ω , linewidth Γ , and $[\text{SSq}]$ as input parameters
- Outputs standard and SCm-modified area eigenvalues, area gap, and phonon correction ratio

6. Conclusion

The SCm extension of the LQG area operator provides a concrete mechanism by which buoyancy-sector phonon modes modulate the quantum geometry of spacetime at the Planck scale. The resulting spectrum retains the discrete character of standard LQG while encoding new physics through $S_{26}^{(3)}$ and $\Phi_{1.25 \text{ THz}}$.

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